

**EXAME
INGLÊS TÉCNICO**

Curso: LECC

Data: 013-08-2021

Turma: LECC11 e LECC12

Duração: 120 min.

Nome do Docente: Anastácio Macuácuá

Pontuação: 230

Nome do aluno _____

Importante: A fraude no exame de uma disciplina tem como consequência a reprovação na disciplina, sem possibilidade do infrator participar no exame de recorrência nem no exame especial (se existir) da disciplina em causa (alínea b, artigo 1 da ADENDA AO RPL).

I. Reading.**General features of operating systems**

- An operating system is a master control program which controls the functions of the computer system as a whole and the running of application programs. All computers do not use the same operating systems. It is therefore important to assess the operating system used on a particular model before initial commitment because some software is only designed to run under the control of specific operating systems. Some operating systems are adopted as 'industry standards' and these are the ones which should be evaluated because they normally have a good software base. The reason for this is that software houses are willing to expand resources on the development of application packages for machines functioning under the control of an operating system which is widely used. The cost of software is likely to be lower in such circumstances as the development costs are spread over a greater number of users, both actual and potential.
- 15 Mainframe computers usually process several application programs concurrently, switching from one to the other, for the purpose of increasing processing productivity. This is known as multiprogramming (multi-tasking in the context of microcomputers), which requires a powerful operating system incorporating work scheduling facilities to control the switching between programs. This entails reading in data for one program while the processor is performing computations on another and printing out results on yet another.
- 25 In multi-user environments an operating system is required to control terminal operations on a shared access basis as only one user can access the system at any moment of time. The operating system allocates control to each terminal in turn. Such systems also require a system for record locking and unlocking, to prevent one user attempting to read a record whilst another user is updating it, for instance. The first user is allocated control to write to a record (or file in some instances) and other users are denied access until the record is updated and unlocked.
- 35 Some environments operate in concurrent batch and real-time mode. This means that a 'background' job deals with routine batch processing whilst the 'foreground' job deals with real-time operations such as airline seat reservations, on-line booking of hotel accommodation, or control of warehouse stocks, etc. The real-time operation has priority, and the operating system interrupts batch processing operations to deal with real-time enquiries or file updates. The stage of batch processing attained at the time of the interrupt is temporarily transferred to backing storage. After the real-time operation has been dealt with, the interrupted program is transferred back to internal memory from backing storage, and processing recommences from a 'restart' point. The operating system also copies to disk backing storage the state of the real-time system every few minutes (periodic check points) to provide a means of 'recovering' the system in the event of a malfunction.
- 45 An operating system is stored on disk and has to be booted into the internal memory (RAM) where it must reside throughout processing so that commands are instantly available. The operating system commands may exceed the internal memory capacity of the computer in which case only that portion of the OS which is frequently used is retained internally, other modules being read in from disk as required. Many microcomputers function under the control of a disk operating system known as DOS.
- 50

1. Answer these questions about the text. (45pts)

1.1. Why is it important to assess the operating system on a computer before buying it?

1.2. What is multiprogramming?

1.3. Fill in the blanks with the words given, sometimes more than one may apply.

Execute – monitor - format – diagnose

1.4. A typical operating system will:

1.5. _____ input and output devices

1.6. _____ The status of hardware devices

1.7. _____ Hardware interrupts

1.8. _____ New disks

1.9. _____ Disk directories

1.10. _____ Disk reading and writing operations

1.11. _____ Disk errors

1.12. _____ Disk commands relating to the deletion, copying, renaming, and dumping of files.

2. Read this passage about the structure of the processor and fill in the gaps using the words below. (45 pts)

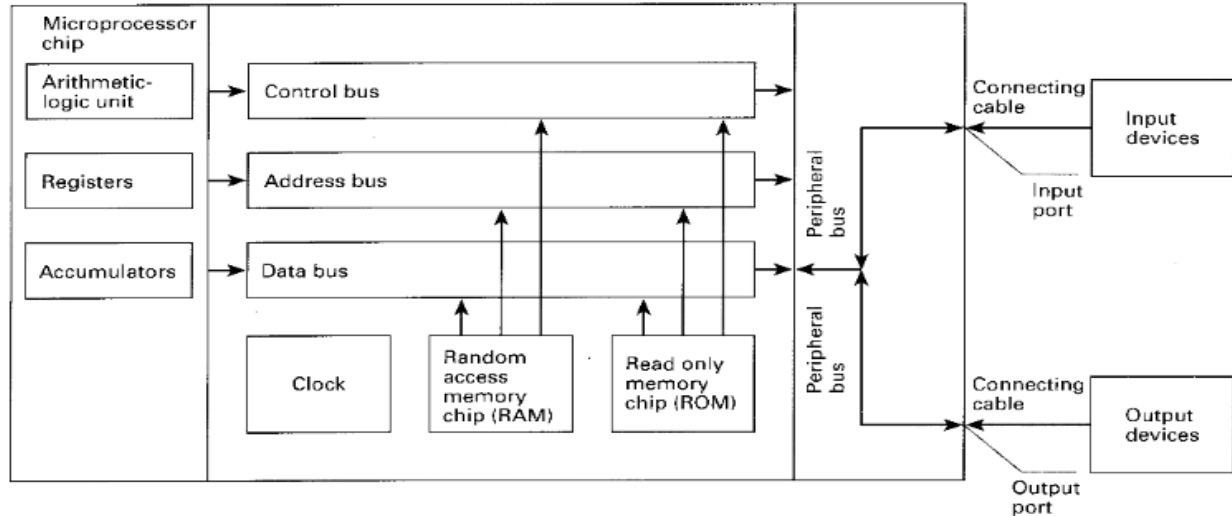
Adaptor boards	registers	microprocessor	clock	conductive
buses	system board	accumulators	input or input devices	



Structure of the processor

The processor consists of a ____1____ which is a circuit board on which are mounted ____2____ chips, memory chips, and other components linked together by ____3____ lines or channels in the form of control, address, and data ____4____. In addition, a processor has ____5____, which are electronic circuits providing specialized functions such as graphics, or which a system board to ____6____. The system board also consists of electronic devices, such as an electronic ____7____ for controlling the speed of operation; ____8____, which store numeric data during the course of processing; and various ____9____, including sequence control register, address register, and function register.

3. Interpret the diagram below using appropriate language and process description. (30 pts)



4. Given the General topics of the paragraphs, write the details related to the structure of an academic paragraph basing predictions of how computers are changing the areas in question.

Paragraph 1 (35 pts)

Computers in Education

Topic Sentence _____

Supporting idea 1 _____

Supporting Idea 2 _____

Argument 1 _____

Argument 2 _____

Paragraph 2 (35 pts)

Computers in medicine

Topic Sentence _____

Supporting idea 1 _____

Supporting Idea 2 _____

Argument 1 _____

Argument 2 _____

5. In no more than 250 words, write an **academic paragraph** talking about the cons and pros of Artificial intelligence. (40 pts)

Good Job